

System Requirements Specification (SRS)

Project: Labaid InsureTech Platform

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1. Introduction

1.1 Purpose

This SRS documents detailed system-level requirements for the Labaid InsureTech platform (mobile apps, partner portals, admin portal, and backend services). Its purpose is to provide an unambiguous specification for design, development, integration, testing, deployment and operations teams.

1.2 Scope

The system will enable digital onboarding, product discovery, policy purchase, digital KYC, payment processing, claims submission and tracking, partner integrations, and admin/insurer workflows. This SRS covers Phase 1 (core digital capabilities) while flagging Phase 2/3 enhancements (AI underwriting, IoT) for future releases. See Business Plan for strategic context.

1.3 Definitions, Acronyms & Abbreviations

- IDRA — Insurance Development & Regulatory Authority (Bangladesh)
 - KYC — Know Your Customer
 - MFS — Mobile Financial Service (bKash, Nagad)
 - API — Application Programming Interface
 - EHR — Electronic Health Record
 - ZHCT — Zero Human Touch Claims
 - UBI — Usage Based Insurance
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2. Overall Description

2.1 Product Perspective

The platform is cloud-native, microservices-based, and mobile-first. Components:

- Mobile apps (iOS/Android) — Customer experience (screens & flows per uploaded designs).
- Partner/Agent Portal — onboarding & embedded flows.
- Admin Portal — product, pricing, user and claims management.

- Back-end Services — authentication, policy engine, payment gateway adapters, document store, notification service, analytics.
- Integrations — Insurer APIs, LabAid systems, MFS/telco APIs. Flow reference: userflow diagram.

2.2 User Classes & Characteristics

- Customers: Mobile-first, diverse digital literacy (urban youth → rural farmers).
- Agent/Partners: Staff at partner organisations (MFS, hospitals, e-commerce).
- Insurer Underwriters: Internal to partner insurers (API consumers).
- Admin: Business admins (policy management, product updates).
- Support/Call Centre: Assisted onboarding & claims help.

2.3 Operating Environment

- Cloud: AWS/Azure preferred; region: Bangladesh (or compliant region).
- Mobile: iOS 13+/Android 9+.
- Browsers for portals: Chrome, Firefox, Edge (latest two versions).

3. System Features & Functional Requirements

Each requirement below has a unique ID (e.g., FR-1). Priority levels: M = Mandatory (Phase 1), D = Desirable (Phase 1), F = Future (Phase 2/3).

3.1 Authentication & User Management

FR-1 (M) — Account Registration: Support phone-based registration with OTP validation (SMS) and capture minimal profile fields (name, DOB, phone, email optional).

FR-2 (M) — Login: OTP login and password-based login; session management and refresh tokens.

FR-3 (M) — Profile Management: Update personal info, nominee details, and document uploads.

FR-4 (M) — Duplicate Prevention: Block duplicate accounts by national ID/phone; provide merge or support flow.

3.2 Digital KYC & Document Verification

FR-5 (M) — Document Upload: Capture images/PDFs for NID/passport, photos, medical docs. UI flows per screens.

FR-6 (M) — OCR & Validation: Extract NID/passport fields via OCR and pre-validate (format + checksum).

FR-7 (D) — eKYC Integration: Integrate with third-party eKYC services for automated verification (Phase 1 if available).

3.3 Product Catalog & Policy Discovery

FR-8 (M) — Product Catalog API: Serve product list with metadata (name, coverage, premiums, insurer, T&Cs).

FR-9 (M) — Filter & Compare: Support filtering (category, premium, coverage) and side-by-side comparison (up to 3). UI screens show these flows.

FR-10 (M) — Product Detail: Full policy wording, exclusions, and illustrative premium breakdown.

3.4 Policy Purchase & Issuance

FR-11 (M) — Multi-step Purchase Flow: Personal details, nominee, document upload, review, payment, confirmation. Flow follows provided mockups.

FR-12 (M) — Premium Calculation: Request insurer API or local pricing engine and show breakdown.

FR-13 (M) — Payment Integration: Support bKash, Nagad, card (PCI-DSS compliant), bank transfer.

FR-14 (M) — Policy Issuance: On successful payment, generate a digital policy certificate (PDF) and store in user account.

FR-15 (D) — Promo/Discounts: Support coupon codes and partner-discount flows.

3.5 Claims Management

FR-16 (M) — Claim Initiation: Policy prefill, claim reason selection, document upload (images, bills), and submission. UI Claim screen referenced.

FR-17 (M) — Status Tracking: Provide claim status updates and admin notes to user.

FR-18 (M) — Admin Workflow: Claims dashboard for triage, verification, approval, rejection, and payment initiation.

FR-19 (D) — Automated Triage: OCR, image verification and rule-based auto-accept for small claims (Phase 2 target).

3.6 Policy Management & Renewals

FR-20 (M) — Policy Dashboard: Active & past policies, download documents, renewal prompts. (UI page: Policy History).

FR-21 (M) — Renewals: Auto-renew option and manual renew flows, with reminders.

FR-22 (D) — Partial Adjustments: Allow address, nominee change requests.

3.7 Notifications & Communication

FR-23 (M) — Notification Engine: Trigger SMS/Push/Email for OTP, purchase confirmation, claims updates, renewal reminders.

FR-24 (D) — Marketing Opt-in/Opt-out: Manage user preferences.

3.8 Admin & Reporting

FR-25 (M) — Admin Portal: User management, product management, claim dashboards, and role-based access control.

FR-26 (M) — Reports: Daily sales, claims ratio, partner performance, policy counts and KPIs (aligned to business plan targets).

3.9 Partner / Agent Portal

FR-27 (M) — Embedded Flow: Partner can initiate policy purchase for end-customer via API or partner portal.

FR-28 (M) — Partner Dashboard: Commission statements, leads, onboarding analytics.

3.10 Audit & Logging

FR-29 (M) — Audit Trail: Immutable logs for critical actions (policy issue, claim approval, payment).

FR-30 (M) — Data Retention: Maintain records to meet regulatory retention periods (configurable).

4. External Interface Requirements

4.1 User Interfaces

- Mobile app screens consistent with provided prototype (purchase, review, payment, claim, history).
- Web admin/partner portals modern UI (desktop-first).

4.2 APIs

- RESTful JSON APIs for: Authentication, Product, Quote/Pricing, Policy issuance, Claim submission, Payment callbacks, Notifications.
- OAuth2 for API client authentication between partners and insurers.

4.3 Third-Party Systems

- Payment Gateways: bKash, Nagad, Card processor (PCI scope).
- Insurer APIs: Premium/underwriting/policy issuance.
- LabAid EHR System: For cashless verification (IPD) and pre-auth (secured HL7/FHIR endpoint or APIs).
- SMS gateway provider for OTP & alerts.

5. Non-Functional Requirements (NFR)

5.1 Reliability & Availability

NFR-1 — Availability: 99.5% uptime (business-critical services)

NFR-2 — Disaster Recovery: RTO < 15 minutes for critical services, RPO < 1 hour.

5.2 Performance

NFR-3 — API response time: median < 200 ms; 95th percentile < 1s.

NFR-4 — App startup time: < 3 seconds on target devices.

5.3 Scalability

NFR-5 — Support up to 1 million registered users and 200k active policies in the initial 24 months with auto-scaling.

5.4 Security

NFR-6 — Encryption: TLS for transit (1.3), AES-256 for data at rest.

NFR-7 — Secrets management & key rotation.

NFR-8 — Admin MFA and RBAC.

NFR-9 — PCI-DSS compliance for card flows.

NFR-10 — Regular penetration tests & security audits.

5.5 Compliance

NFR-11 — IDRA reporting capabilities and data formats.

NFR-12 — AML/CFT detection hooks and SAR reporting.

5.6 Usability

NFR-13 — Mobile-first UX, accessible flows for low-literacy users (voice-help optional).

NFR-14 — Multi-language support (Bengali + English).

6. Data Model & Storage Requirements

6.1 Data Entities (minimum)

- User (UserID, name, phone, email, NID, DOB, KYC status)
- Policy (PolicyID, productID, insured period, premium, status, insurer reference)
- Claim (ClaimID, policyID, date, documents, status, admin notes)
- Product (productID, coverage, T&Cs, premium table)
- Transaction (TransactionID, amount, gateway, status)
- AuditLog (event, user, timestamp)

6.2 Storage & Retention

- Document store (S3 or equivalent) with encryption, immutable storage for KYC docs.
- Relational DB (Postgres) for core entities; Data warehouse (Redshift/BigQuery) for analytics.
- Retention policy: minimum 7 years for policy & claim records (adaptable to IDRA requirements).

7. Security & Compliance Requirements

7.1 Authentication & Authorization

- JWT or OAuth2 for sessions.
- Role-based access control for portals.

7.2 Data Protection

- PII encrypted at rest.
- Masking of NID in logs and limited access to raw documents.

7.3 Vulnerability Management

- Quarterly vulnerability scans.
- Annual third-party penetration testing.

7.4 Regulatory Measures

- KYC/eKYC flow that satisfies IDRA.
- AML rule engine with flagged escalation to compliance team.
- Audit logs and reporting to meet IDRA periodic submissions.

8. Performance & Scalability Requirements

- Peak TPS (Transactions per second) for purchase flow: baseline 50 TPS, scale to 500 TPS via auto-scaling.
- Concurrent active sessions target: 100k with <5% degradation.
- File upload: support 10 MB images with background upload/resume.

9. Operational Requirements & Support

- 24x7 monitoring with alerts (PagerDuty).
- Centralized logging (ELK/Datadog) and metrics.
- Runbooks for incident management.
- Support SLA: 1 hour for P1 incidents, 4 hours for P2.

10. Acceptance Criteria & Test Summary

10.1 Acceptance Criteria (high-level)

- FR-critical workflows (Registration → Purchase → Payment → Policy Issuance → Claim Submission) operate end-to-end in the staging environment without manual intervention for standard products.
- Payment callback reliability > 99.9%.
- OTP deliverability > 98% and verification success > 95%.
- Policy issuance within 30 seconds for auto-issuance products.

10.2 Testing Types

- Unit & integration tests for all services.
- End-to-end QA (customer flows) with both manual and automated scripts.
- Performance/load testing to validate NFRs.
- Security testing (SAST/DAST, pen-testing).
- UAT with LabAid internal pilot users (employee cohort) prior to market launch.

11. Traceability Matrix & Change Control

11.1 Traceability

All system requirements (FR-/NFR-) will be linked to BRD objectives and business KPIs (policy targets, claims SLA) in the project tracking tool (JIRA). Example mapping:

- Business Objective: “Digital onboarding & 40,000 policies in 2026” → FR-1, FR-11, FR-13, FR-20.

11.2 Change Control

- Changes must be submitted via RFC with business justification.
- Impact analysis (cost/time/Risk) required.
- Steering committee approval for scope changes affecting timeline or budget.

12. Appendices

A. Use Cases (selected)

- UC-01: Register & complete profile (happy path & exceptions).
IT01_Insuretech Userflow Diagram
- UC-02: Search & purchase Device Insurance (includes document upload & payment).
IT02_Insuretech Proposed Design...
- UC-03: Initiate claim with supporting documents & track status.
IT02_Insuretech Proposed Design...

B. Screens & Flows (references)

- User Flow Diagram (Registration → Purchase → Claim).
IT01_Insuretech Userflow Diagram
- Proposed App UI/UX (Insucare v1 screens: purchase flows, payment, policy history, claim UI).
IT02_Insuretech Proposed Design...

C. Cross-references to Business Plan

- Financial targets, milestone timelines and strategic capabilities (AI underwriting, partner integrations) are defined in the Business Plan and are used to prioritise Phase 1 and roadmapping for Phase 2.
Working_Paper_V0
Working_Paper_V1

Sign-off

By signing below stakeholders accept this SRS as the authoritative technical specification for Phase 1 development of the Labaid InsureTech platform.

- Product Owner / Labaid InsureTech: _____ Date: _____
 - CTO / Vendor Lead: _____ Date: _____
 - Compliance / Legal: _____ Date: _____
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